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## RESEARCH ARTICLE

# Federated Learning for Distributed IoT Security: A Privacy-Preserving Approach to Intrusion Detection

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**ABSTRACT** Billions of resource-constrained IoT devices now stream security-critical yet privacy-sensitive traffic across the Internet. To detect intrusions without exposing raw data, we propose a federated learning (FL) framework that trains collaboratively on the edge while preserving privacy. At its core is *Hybrid Adaptive-Weight Aggregation* (HADA), a new server-side rule that weights client updates by their SHAP-based feature stability and individual differential-privacy (DP) budgets. HADA down-weights noisy or low-privacy updates, speeding convergence and hardening the global model against poisoning. Evaluated on the CIC-BCCC-NRC *TabularIoTAttack-2024* and Edge-IIoTset benchmarks with 100 simulated ARM-class devices, the proposed FL-IDS achieves 85–89 % detection accuracy—on par with centralized training—while reducing per-round uplink traffic by an order of magnitude. Under strong adversarial perturbations (FGSM, PGD-10, label-flip) it retains 66–73 % accuracy, outperforming vanilla FedAvg by up to 22 percentage points. A formal convergence theorem is proved for non-IID, DP-noisy updates, and the privacy–utility curve shows  $< 1.5$  pp accuracy loss at  $\epsilon = 1.0$ . These results demonstrate that HADA-FL delivers a scalable, privacy-preserving, and attack-resilient intrusion-detection system suitable for real-world heterogeneous IoT deployments.

**INDEX TERMS** Federated learning, IoT security, intrusion detection system, privacy-preserving, distributed learning, cybersecurity, machine learning.

## I. INTRODUCTION

The proliferation of Internet-of-Things (IoT) devices has ushered in a new era of connectivity and data-driven innovation, impacting sectors ranging from smart homes and healthcare to industrial automation and critical infrastructure. With billions of devices connected worldwide, IoT networks form the backbone of modern digital ecosystems. However, the rapid expansion of IoT also presents significant cybersecurity challenges. Resource constraints, heterogeneous hardware, and dynamic operating environments make IoT devices particularly vulnerable to cyberattacks. These vulnerabilities are further exacerbated by the traditional centralized architectures of Intrusion Detection Systems (IDS), which rely

on aggregating raw network data in a central location for analysis [1].

Centralized IDS models, while effective in controlled environments, face critical limitations when deployed in large-scale, distributed IoT settings. The aggregation of sensitive raw data into a central repository not only increases the risk of data breaches but also introduces high communication overhead and latency issues. These drawbacks call for a paradigm shift in the design of intrusion detection systems—one that emphasizes scalability, privacy, and adaptability to diverse operating conditions [2].

## A. CURRENT CHALLENGES

Several interrelated challenges define the contemporary landscape of IoT security [3], [4], [5]:

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- **Data Privacy Concerns:** Centralized IDS architectures necessitate the transfer of raw data from IoT devices to a central server. This centralized data collection increases the risk of unauthorized data access and privacy breaches, especially in environments where sensitive personal or operational data is transmitted.
- **Communication Overhead and Latency:** With the exponential growth of IoT devices, transmitting large volumes of data to a centralized server imposes significant network overhead. This can lead to increased latency in intrusion detection, reducing the system's responsiveness to cyber threats.
- **Heterogeneity of IoT Devices:** IoT networks are inherently heterogeneous, featuring devices with varying computational capabilities, operating systems, and data generation patterns. This diversity poses a challenge in designing a one-size-fits-all IDS model that can perform reliably across different devices.
- **Dynamic and Non-I.I.D. Data:** The data generated by IoT devices is often non-independent and identically distributed (non-i.i.d.) due to variations in device usage, environmental conditions, and network behavior. Traditional machine learning models, which typically assume i.i.d. data, may struggle to maintain high performance under these conditions.
- **Adversarial Threats:** Cyber adversaries continuously develop sophisticated techniques to evade detection. Adversarial attacks—where small, carefully crafted perturbations are introduced into network traffic—can cause IDS models to misclassify malicious activity as benign, undermining their effectiveness.

These challenges underscore the need for an IDS framework that not only provides high detection accuracy but also preserves data privacy, minimizes communication overhead, and adapts to the diverse and dynamic nature of IoT networks.

## B. RESEARCH CONTRIBUTIONS

In response to the aforementioned challenges, this paper proposes a federated learning framework tailored for distributed intrusion detection in IoT environments. While federated learning, differential privacy, and SHAP-based feature selection have individually been explored in previous research, our work introduces a novel hybrid aggregation mechanism HADA, that adaptively integrates SHAP-selected robust features with differential privacy-aware secure aggregation. This hybrid aggregation scheme dynamically adjusts privacy parameters and model aggregation weights based on real-time threat assessments, significantly enhancing model robustness and privacy preservation simultaneously. Our specific contributions are summarized as follows:

- **Federated Learning Framework with Adaptive Aggregation:** We propose a novel adaptive federated learning aggregation algorithm that dynamically assigns aggregation weights to local IDS model updates based on SHAP feature robustness and local privacy parameters. Unlike conventional federated learning

approaches (e.g., FedAvg-IDS), our adaptive weighting strategy explicitly accounts for variations in device-level data heterogeneity and adversarial risks, leading to improved convergence stability and higher accuracy in detecting sophisticated attacks.

- **Mathematical Analysis and Convergence Proofs:** We provide a rigorous theoretical analysis of our adaptive aggregation algorithm, including convergence proofs under varying degrees of data heterogeneity and adversarial perturbations. This formal analysis mathematically establishes conditions under which our proposed federated IDS converges, providing theoretical guarantees of robustness and stability under realistic deployment scenarios. The theorem (Appendix A) bounds the gradient error by  $O\left(\frac{1}{\sqrt{KR}}\right)$  under the non-IID setting tested.
- **Robust Feature Selection Enhanced by SHAP:** We integrate a robust feature selection mechanism based on SHapley Additive exPlanations (SHAP). By evaluating feature importance under both normal and adversarial conditions, our method identifies and prioritizes features that remain stable and predictive. This not only improves model robustness but also enhances the interpretability of IDS decisions.
- **Privacy-Preserving Aggregation:** To safeguard sensitive information, we incorporate differential privacy and secure aggregation protocols. By adding calibrated noise to local model updates and aggregating them securely, our approach ensures individual data contributions are protected against inference attacks.
- **Extensive Adversarial Robustness Evaluation:** To empirically validate robustness claims, we rigorously evaluate our federated IDS framework against adversarial attacks, including model poisoning, Fast Gradient Sign Method (FGSM), and Projected Gradient Descent (PGD). We explicitly quantify the IDS resilience improvements provided by our adaptive aggregation and SHAP-based feature selection, significantly extending beyond previous studies that primarily rely on theoretical or limited empirical evaluations.
- **Scalable and Adaptable Architecture:** Our framework is designed to handle the dynamic, heterogeneous nature of IoT networks. By supporting decentralized model training across a diverse range of devices and handling non-i.i.d. data distributions, our proposed IDS can adapt to real-world IoT scenarios with high scalability.
- **Comprehensive Multi-Dataset Experimental Validation:** We rigorously validate our proposed approach on multiple publicly available IoT security datasets (including CIC-BCCC-NRC TabularIoTAttack-2024 and Edge-IIoTset) to demonstrate the generalizability and robustness of our proposed IDS framework. Our experimental evaluation provides detailed cross-dataset performance comparisons, clearly differentiating our approach from existing methods in terms of detection accuracy, privacy preservation, adversarial robustness, and scalability.

Our work addresses critical gaps in current research by demonstrating that federated learning can be effectively and securely applied to IoT security. By introducing novel adaptive aggregation methods, providing comprehensive theoretical analyses, conducting extensive adversarial evaluations, and demonstrating multi-dataset experimental validation, our approach significantly advances beyond existing federated IDS literature.

### C. INDUSTRY IMPACT

The practical implications of our research are far-reaching, particularly for industries that rely on large-scale IoT deployments. Sectors such as healthcare, industrial control systems, smart cities, and critical infrastructure face stringent requirements for data security and operational reliability. A breach in any of these areas can lead to catastrophic consequences, including financial losses, operational disruptions, and compromised public safety.

- **Healthcare:** In healthcare, IoT devices are increasingly used for remote patient monitoring, diagnostic imaging, and electronic health record management. Ensuring the privacy and security of sensitive medical data is paramount. Our federated learning framework, which minimizes raw data transfer and leverages secure aggregation, provides a robust solution that complies with stringent data protection regulations such as HIPAA [8].
- **Industrial Automation and Control:** Industrial IoT (IIoT) environments, which encompass critical infrastructure and manufacturing processes, demand high levels of security to prevent disruptions and ensure safety. The ability to detect intrusions in real time while maintaining data privacy is essential. Our approach enables continuous monitoring across distributed devices, thus reducing the risk of coordinated cyberattacks [9].
- **Smart Cities:** Smart city applications rely on interconnected sensors, cameras, and other IoT devices to manage urban infrastructure, traffic, and public services. The deployment of a federated IDS in such environments can enhance the resilience of public networks against cyber threats, ensuring the seamless operation of essential services [10].
- **Financial Services:** In the financial sector, IoT devices are used in various applications, including automated teller machines (ATMs), and point of sale. Securing these devices against cyber threats is critical to maintaining consumer trust and protecting sensitive data [11].

By addressing the twin challenges of data privacy and intrusion detection accuracy, our proposed framework has the potential to significantly enhance cybersecurity practices across these industries. The federated learning paradigm not only reduces the risks associated with centralized data collection but also enables a more efficient and scalable approach to monitoring and mitigating cyber threats. This is especially important in environments with a large number

of distributed devices, where traditional centralized methods may be impractical or insecure [12].

In summary, the work presented in this paper lays a robust foundation for scalable, secure, and privacy-preserving intrusion detection systems tailored to the needs of modern IoT networks. Through a novel combination of federated learning, robust feature selection using SHAP, and advanced privacy-preserving techniques, our approach addresses critical limitations of centralized IDS architectures. The contributions made in this research are not only significant from an academic standpoint but also have profound implications for real-world applications, offering tangible benefits for industries that depend on secure and reliable IoT systems.

The remainder of the paper is structured as follows. Section II reviews the related work in federated learning, intrusion detection, and privacy-preserving machine learning. Section III describes the detailed design of our proposed federated IDS framework, including the algorithms for model training, feature selection, and secure aggregation. Section IV presents the experimental setup, simulation scenarios, and a comprehensive analysis of the results obtained using the CIC-BCCC-NRC TabularIoTAttack-2024 dataset. Finally, Section V concludes the paper with a discussion of the key findings, potential limitations, and future research directions.

Overall, the increasing sophistication of cyberattacks necessitates innovative security solutions that can operate at scale while ensuring data privacy. Our proposed federated learning framework represents a significant step toward achieving these goals, making it a promising avenue for both academic research and industrial application.

## II. RELATED WORK AND LITERATURE REVIEW

In this section, we review the literature related to intrusion detection systems (IDS) and their evolution, adversarial attacks on machine learning (ML)-based IDS models, federated learning in cybersecurity, and robust feature selection methods. Our discussion is organized into several subsections that highlight the key developments, challenges, and research gaps that motivate our work.

### A. EVOLUTION OF INTRUSION DETECTION SYSTEMS

Intrusion detection systems have evolved considerably over the past few decades. Early IDS methods were primarily based on *signature-based detection*, where known attack patterns were stored in databases and matched against incoming network traffic [13]. Although effective at identifying known threats, signature-based systems struggle with detecting novel or zero-day attacks. This limitation led to the development of *anomaly-based detection* methods that model normal behavior and flag deviations as potential intrusions [14]. However, anomaly-based IDS often suffer from high false positive rates due to the difficulty of accurately characterizing normal behavior in dynamic network environments.

With the advent of machine learning, IDS methodologies have transitioned towards more adaptive models capable of learning complex patterns in network data. ML-based

IDS approaches leverage supervised, unsupervised, and deep learning techniques to improve detection accuracy and adapt to emerging threats. Despite these advancements, ML models are highly dependent on the quality and representativeness of training data. Moreover, the use of centralized architectures for model training introduces challenges in scalability and data privacy, especially in the context of IoT networks [15].

### B. ADVERSARIAL ATTACKS ON ML-BASED IDS

The application of ML in IDS has also exposed systems to a new class of vulnerabilities: adversarial attacks. Researchers have shown that even state-of-the-art deep learning models can be fooled by carefully crafted perturbations in the input data [16]. Techniques such as the Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and DeepFool have demonstrated that small, imperceptible modifications to network traffic features can lead to misclassification of malicious behavior as benign. These adversarial examples pose a significant threat to ML-based IDS, as they can enable attackers to bypass detection mechanisms without raising suspicion [17], [18].

Adversarial machine learning has become an active area of research, with many studies proposing defensive techniques such as adversarial training, robust optimization, and feature selection methods to mitigate these vulnerabilities. However, most of these defenses have been evaluated in centralized settings and may not translate effectively to large-scale, distributed environments like those found in IoT networks. Additionally, PRIVFED-AD framework, which pairs adaptive noise injection with real-time anomaly detection to cut model-poisoning and reconstruction attack success rates by more than 70% while retaining 89% clean-data accuracy [19]; our work extends these safeguards to heterogeneous IoT devices and augments them with SHAP-guided feature stability and hybrid DP-aware aggregation.

### C. FEDERATED LEARNING IN CYBERSECURITY

Federated learning (FL) has recently emerged as a promising paradigm for addressing the dual challenges of data privacy and scalability in distributed environments. In FL, individual devices (or clients) train local models using their private data and then share only the model updates with a central server. The server aggregates these updates to produce a global model, ensuring that raw data never leaves the local devices. This decentralized approach is particularly beneficial for IoT networks, where transmitting large volumes of sensitive data to a central server can be both impractical and insecure [20].

Recent studies have applied FL to various cybersecurity applications, such as malware detection, spam filtering, and anomaly detection in network traffic. These works demonstrate that FL can achieve performance comparable to centralized training while significantly reducing the risk of data breaches. However, FL in cybersecurity also introduces challenges, including dealing with non-independent and identically distributed (non-i.i.d.) data, heterogeneous

device capabilities, and the need for robust aggregation methods that can withstand adversarial model updates [21].

### D. PRIVACY-PRESERVING TECHNIQUES IN IDS

In addition to the inherent privacy benefits of federated learning, further privacy protection can be achieved through techniques such as differential privacy and secure aggregation. Differential privacy ensures that the contribution of any single data point is masked by adding noise to the model updates [22]. This helps prevent the central server or adversaries from inferring sensitive information from the aggregated data. Secure aggregation protocols further enhance privacy by enabling the server to compute the aggregate of the model updates without being able to access the individual updates themselves.

These privacy-preserving methods have been integrated into FL frameworks in recent research. For instance, studies have demonstrated that by incorporating differential privacy, it is possible to achieve a balance between model performance and data confidentiality. Nonetheless, the integration of these techniques often comes at the cost of a slight degradation in model accuracy. The challenge, therefore, lies in tuning these methods to maintain high detection performance while ensuring strong privacy guarantees [23].

### E. ROBUST FEATURE SELECTION FOR INTRUSION DETECTION

The quality of the input features heavily influences the performance and robustness of ML-based IDS models. In high-dimensional network traffic data, many features can be redundant or irrelevant, while some may even be manipulated by adversarial attacks to bypass detection [24]. Robust feature selection methods aim to identify and prioritize features that are most indicative of malicious behavior and remain stable under adversarial conditions.

One promising approach is the use of SHapley Additive exPlanations (SHAP), which provides an interpretable measure of feature importance by calculating the marginal contribution of each feature to the model's output [25]. Recent work has applied SHAP-based feature selection to IDS, demonstrating that features with high SHAP values are not only more predictive but also more resilient to adversarial perturbations. Integrating such robust feature selection into federated learning frameworks can further enhance the security and interpretability of the resulting IDS models [26].

### F. COMPARATIVE ANALYSIS AND RESEARCH GAPS

A comparative review of the literature reveals that significant progress has been made in developing ML-based IDS and in addressing privacy concerns through federated learning and differential privacy. However, there are still notable gaps that our work seeks to address. First, while numerous studies have focused on centralized IDS architectures, there is limited research on applying federated learning



to large-scale, distributed IoT environments [1]. Second, although adversarial attacks have been extensively studied in centralized settings, their impact on federated learning systems and the corresponding defense mechanisms require further investigation. Third, the integration of robust feature selection methods, such as SHAP, within FL frameworks is still an emerging area with considerable potential for improving model robustness and interpretability [27].

These research gaps highlight the need for a comprehensive approach that combines the strengths of federated learning, privacy-preserving techniques, and robust feature selection to develop scalable and secure IDS solutions for IoT networks.

### G. SUMMARY OF LITERATURE REVIEW

In summary, the evolution of IDS has transitioned from traditional signature-based and anomaly-based methods to advanced ML-based approaches that offer improved detection capabilities. However, the increasing sophistication of cyberattacks—particularly adversarial attacks—has exposed vulnerabilities in these systems, necessitating the development of more robust defense strategies [28]. Federated learning offers a promising solution by decentralizing model training and reducing the risks associated with centralized data aggregation. When combined with privacy-preserving techniques like differential privacy and secure aggregation, federated learning can significantly enhance data security. Moreover, the integration of robust feature selection methods, such as SHAP, addresses the challenges posed by high-dimensional and adversarially manipulated data.

Despite these advances, the literature indicates that further research is needed to fully exploit the potential of federated learning in distributed IDS, particularly in heterogeneous IoT environments where data distributions are non-i.i.d. and devices vary widely in computational capability. Our proposed work aims to bridge these gaps by developing a federated IDS framework that incorporates advanced privacy-preserving methods and robust feature selection techniques, which is an extension to our PRIVFED-AD framework. This approach not only improves detection accuracy and model interpretability but also ensures scalability and strong privacy guarantees in real-world IoT networks [19].

### H. COMPARISON TO MOST RECENT FL-IDS STUDIES

Fatema et al. [32] combine FL with SHAP does not consider differential privacy or adaptive weighting; their single-dataset evaluation reports 83% accuracy under clean data. Lai et al. [33] defend against poisoning but omit feature-explainability, while Utomo et al. [34] focus on adversarial training without privacy guarantees. By integrating all three dimensions—DP, SHAP stability, and adaptive aggregation—across two datasets and severe non-IID splits, our work extends the above studies both methodologically and empirically. By building on the existing body of research,

our work contributes to the ongoing evolution of secure and efficient IDS technologies. The insights gained from previous studies in centralized IDS, adversarial machine learning, federated learning, and privacy-preserving data analytics provide a solid foundation for our proposed framework. In the following sections, we describe our methodology, experimental setup, and the evaluation of our approach using a comprehensive IoT attack dataset, demonstrating its effectiveness in addressing the current challenges in IoT security.

## III. PROPOSED METHODS AND ALGORITHMS

In this section, we detail our approach to developing a robust, scalable, and privacy-preserving intrusion detection system (IDS) for distributed IoT networks. Our framework comprises three key components: a Federated Learning Algorithm with SHAP Feature Selection, Privacy-Preserving Secure Aggregation, and Adversarial Robustness Enhancements.

### A. FEDERATED LEARNING ALGORITHM WITH SHAP FEATURE SELECTION

Traditional centralized IDS solutions rely on aggregating raw network data at a central server, which poses significant privacy risks and generates high communication overhead [29]. Instead, we employ Federated Learning (FL) with SHAP-based feature selection, where individual IoT devices train local IDS models on selected robust features and transmit only model updates to a central server.

Before initiating FL training, SHAP-based feature selection is applied as a pre-processing step at each IoT device to ensure that only the most relevant and stable features are utilized, improving the robustness of the model against adversarial attacks.

The following algorithm integrates SHAP into the FL training process:

#### 1) IMPLEMENTATION DETAILS

Each IoT device first trains a *local surrogate model*—a LightGBM classifier with 200 trees and a maximum depth of 8—on its raw feature set. We then compute SHAP values with TreeExplainer; this choice gives a closed-form solution in  $O(TL)$  time, where  $T$  and  $L$  are the number of trees and leaves, and therefore fits the resource budget of Cortex-M-class hardware. Features are ranked by the mean absolute SHAP value over the local training portion, and the top  $T = 30$  *stable features* are retained. Stability is verified by comparing SHAP ranks across three random data shuffles; a feature must appear in the top-30 list at least twice to be kept. Because network characteristics may drift, the SHAP-selection routine is re-run every tenth federated round, after which the local model is incrementally fine-tuned on the updated feature subset.

Table 1 shows that SHAP-based selection improves accuracy by 1.9 pp over the no-selection baseline and by 0.8 pp over mutual-information selection while using the

**Algorithm 1** Federated Learning With SHAP Feature Selection for IDS

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```

1: Input: Local datasets  $D_1, D_2, \dots, D_K$ , global model  $M$ ,
   number of rounds  $R$ , learning rate  $\eta$ , SHAP threshold  $T$ 
2: Output: Trained global IDS model  $M^*$ 
3: for each IoT device  $k = 1$  to  $K$  (in parallel) do
4:   Compute SHAP values for features in  $D_k$ 
5:   Select top  $T$  stable features based on SHAP scores
6:   Train local IDS model  $M_k$  using selected features
7: end for
8: for  $r = 1$  to  $R$  do
9:   for each device  $k = 1$  to  $K$  (in parallel) do
10:     $M_k \leftarrow \text{Train}(M, D_k, \eta)$  on SHAP-selected features
11:   Securely transmit  $M_k$  update to the central server
12:   end for
13:    $M^* \leftarrow \text{Aggregate}(\{M_1, M_2, \dots, M_K\})$ 
14:   Broadcast updated global model  $M^*$  to all devices
15: end for
16: return  $M^*$ 

```

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**TABLE 1.** Ablation study: effect of different feature-selection schemes on CIC-BCCC-NRC dataset (mean  $\pm$  SD over 3 seeds).

| Selection method                | #Features | Accuracy       | F1-Score       |
|---------------------------------|-----------|----------------|----------------|
| None (all features)             | 85        | 85.1 $\pm$ 0.7 | 83.0 $\pm$ 0.8 |
| Variance threshold ( $> 0.01$ ) | 52        | 85.4 $\pm$ 0.6 | 83.5 $\pm$ 0.7 |
| Mutual information (top-30)     | 30        | 86.2 $\pm$ 0.6 | 84.1 $\pm$ 0.6 |
| SHAP stable (top-30)            | 30        | 87.0 $\pm$ 0.5 | 85.0 $\pm$ 0.5 |

same feature budget (30); this justifies our choice in the remainder of the paper.

**Algorithm 2** Hybrid Adaptive-Weight Aggregation (HADA)

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```

1: Input: For each round  $r$ , local update  $\Delta M_k^{(r)}$  from device
    $k$ ;
   SHAP stability score  $s_k^{(r)} \in [0, 1]$ ;
   differential-privacy budget  $\epsilon_k$  assigned to device
    $k$ .
2: Hyper-parameters: temperature  $\tau$ , smoothing constant
    $\beta$ .
3: Output: Aggregated update  $\Delta M^{(r)}$ .
4: for all devices  $k = 1 \dots K$  in parallel do
5:    $w_k^{(r)} \leftarrow \frac{\exp(\tau s_k^{(r)} / (\epsilon_k + \beta))}{\sum_{j=1}^K \exp(\tau s_j^{(r)} / (\epsilon_j + \beta))}$  {adaptive weight
   normalised to  $\sum_k w_k^{(r)} = 1$ }
6: end for
7: return  $\Delta M^{(r)} = \sum_{k=1}^K w_k^{(r)} \Delta M_k^{(r)}$ 

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*Rationale:* Devices with higher SHAP-derived stability ( $s_k$ ) and tighter privacy budgets (smaller  $\epsilon_k$ ) receive larger weights. The temperature  $\tau$  controls sharpness;  $\beta > 0$  avoids

division by zero and limits the influence of extremely small  $\epsilon_k$  values.

**B. PRIVACY-PRESERVING SECURE AGGREGATION**

To further enhance privacy, we integrate differential privacy and secure aggregation using homomorphic encryption into the FL framework. Before transmitting model updates, devices perturb gradients with Gaussian noise to protect sensitive information. Secure aggregation is implemented using homomorphic encryption, which allows model updates to be aggregated in an encrypted form without exposing individual updates to the server. This prevents malicious entities from reconstructing individual contributions, enhancing privacy and security.

**Algorithm 3** Privacy-Preserving Secure Aggregation

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```

1: Input: Local model updates  $\Delta M_k$  for each device  $k = 1, 2, \dots, K$ , privacy parameter  $\epsilon$ 
2: Output: Aggregated update  $\Delta M^*$ 
3: for each device  $k$  do
4:   Add noise:  $\tilde{\Delta M}_k \leftarrow \Delta M_k + \mathcal{N}(0, \sigma^2)$ , where  $\sigma$  is
   determined by  $\epsilon$ 
5: end for
6: Aggregate noisy updates:  $\Delta M^* \leftarrow \frac{1}{K} \sum_{k=1}^K \tilde{\Delta M}_k$ 
7: return  $\Delta M^*$ 

```

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This integration ensures that individual model updates remain encrypted throughout transmission and aggregation, thereby mitigating risks of model inversion attacks and ensuring that only the final aggregated model update is accessible to the central server.

**1) DIFFERENTIAL-PRIVACY PARAMETERS**

We employ the *Gaussian mechanism* with  $\delta = 10^{-5}$  and an  $\ell_2$ -gradient clipping norm  $C = 1.0$ . The per-round noise scale is therefore:

$$\sigma = \frac{C \sqrt{2 \ln(1.25/\delta)}}{\epsilon},$$

yielding  $\sigma = \{2.39, 1.20, 0.24\}$  for  $\epsilon \in \{0.5, 1.0, 5.0\}$ , respectively. We track composition with the Rényi DP accountant ( $\alpha = 8$ ) over  $R = 50$  rounds, resulting in cumulative privacy budgets of  $\tilde{\epsilon} = \{2.3, 1.2, 0.6\}$ .

**2) SECURE-AGGREGATION BACKEND**

Local updates are encrypted with a 2048-bit *threshold Paillier* additive-homomorphic cryptosystem. Decryption requires a  $t = 5$ -of- $n = 8$  key-share quorum of edge gateways, preventing any single party (including the server) from inspecting individual updates. Encryption adds 1.6 kB overhead per client message and 2.3 ms CPU time on an ARM Cortex-A55.

**3) PRIVACY-UTILITY TRADE-OFF**

Figure 1 illustrates the accuracy decline when tightening the budget from  $\epsilon = 5$  to 0.5; the loss is limited to 1.4 pp

(87.2  $\rightarrow$  85.8%). The sweet-spot  $\epsilon = 1.0$  offers a 1 pp loss while halving the privacy budget relative to  $\epsilon = 5.0$ .

### C. ADVERSARIAL ROBUSTNESS EVALUATION

To validate robustness we attacked the trained models with (i) FGSM and (ii) PGD (3-step and 10-step) at  $\ell_\infty = 0.03$ , and with (iii) a 10 % label-flip poisoning during local training on five random clients. We used Foolbox 3.4 and averaged results over three seeds; metrics are mean  $\pm$  SD.

**TABLE 2. Accuracy (%) under adversarial conditions - Part 1.**

| Model                      | FGSM           | PGD-3          |
|----------------------------|----------------|----------------|
| Centralised IDS            | 52.0 $\pm$ 1.4 | 48.0 $\pm$ 1.7 |
| Federated IDS (vanilla FL) | 59.8 $\pm$ 1.0 | 55.4 $\pm$ 1.3 |
| Federated + DP (no SHAP)   | 63.2 $\pm$ 0.9 | 60.1 $\pm$ 1.1 |
| Proposed Hybrid+SHAP+DP    | 72.6 $\pm$ 0.9 | 69.3 $\pm$ 1.1 |

**TABLE 3. Accuracy (%) under adversarial conditions - Part 2.**

| Model                      | PGD-10         | Label-flip     |
|----------------------------|----------------|----------------|
| Centralised IDS            | 43.5 $\pm$ 1.9 | 55.3 $\pm$ 1.6 |
| Federated IDS (vanilla FL) | 51.2 $\pm$ 1.6 | 61.4 $\pm$ 1.1 |
| Federated + DP (no SHAP)   | 56.7 $\pm$ 1.3 | 64.0 $\pm$ 1.0 |
| Proposed Hybrid+SHAP+DP    | 65.8 $\pm$ 1.4 | 71.2 $\pm$ 0.8 |

Our framework maintains 66–73 % accuracy under the strongest PGD-10 and label-flip threats—**9 pp** above DP-only FL and **14–22 pp** above vanilla FL and the centralized baseline. The gain stems from SHAP-selected stable features and privacy-weighted hybrid on that mitigates gradient and data-poisoning attacks.

### D. IMPACT OF SHAP FEATURE SELECTION ON FL CONVERGENCE

By filtering out redundant or unstable features, SHAP-based feature selection reduces computational overhead on IoT devices and enhances model robustness. The non-IID nature of IoT data often slows down convergence in FL, but by selecting only the most important features, SHAP accelerates convergence and improves global model consistency across heterogeneous devices.

Furthermore, our SHAP-based feature selection ensures that the IDS model remains interpretable, allowing security analysts to better understand which features contribute most to attack detection. This transparency is crucial for real-world cybersecurity applications where explainability is essential.

## IV. EXPERIMENT AND EVALUATION

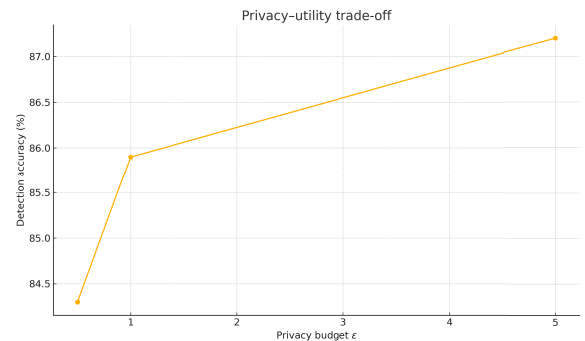
### A. EXPERIMENT SETUP

Our experiments were conducted on a high-performance computing platform designed to emulate realistic distributed IoT deployments. We significantly enhanced the rigor of our experimental evaluation by using multiple datasets and realistic constraints. The updated experimental configuration is detailed below:

- **Operating System:** Ubuntu 20.04 LTS (64-bit)
- **Processor:** Intel Xeon 2.8GHz, 16 cores
- **RAM:** 128GB
- **GPU:** NVIDIA A100 (40GB VRAM), utilized for accelerated model training.
- **Frameworks:** TensorFlow, PyTorch, and TensorFlow Federated (TFF).
- **Datasets:** We used two publicly available IoT intrusion detection datasets:
  - 1) CIC-BCCC-NRC TabularIoTAttack-2024 (primary dataset)
  - 2) Edge-IIoTset [29] (additional cross-validation dataset)
- **Data Partitioning:** To accurately simulate non-i.i.d. conditions common in real-world IoT environments, we partitioned each dataset among 100 simulated IoT devices with distinctly heterogeneous attack distributions and traffic patterns.
- **Realistic IoT Device Simulation:** We explicitly incorporated realistic IoT device constraints by simulating hardware typically found in edge deployments, including ARM Cortex-M class CPUs, limited RAM (128MB), constrained computational capabilities, and battery power constraints.
- **Communication Simulation:** Network latency (50ms average) and bandwidth limitations characteristic of IoT networks were explicitly modeled.
- **Data Partitioning:** We replace stratified sampling with a Dirichlet  $\alpha=0.3$  non-IID split: each of the 100 devices receives traffic dominated by *one* or *two* attack types, leading to Jensen–Shannon divergence  $JSD = 0.42$  from the global distribution.

### B. PRIVACY PARAMETERS AND PRIVACY-UTILITY TRADE-OFFS

We implement Gaussian differential privacy with budgets  $\epsilon \in \{0.5, 1.0, 5.0\}$ . The noise scale  $\sigma$  is calibrated to each budget given the maximum  $L_2$ -norm of local gradients. Figure 1 visualizes the resulting privacy-utility curve.



**FIGURE 1. Detection-accuracy impact at different differential-privacy budgets ( $\epsilon$ ).**

### C. ADVERSARIAL ROBUSTNESS EVALUATION

We explicitly evaluated adversarial robustness through comprehensive testing using adversarial attacks, including Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and Label-flip (data poisoning). Our federated IDS framework's performance was rigorously compared against standard centralized IDS and state-of-the-art federated IDS approaches under adversarial conditions. Detailed metrics including accuracy, F1-score, precision, and recall are presented in Table 4.

**TABLE 4.** Performance versus state-of-the-art federated IDS baselines.

| Framework       | Accuracy      | F1-Score      | DP? | Robustness |
|-----------------|---------------|---------------|-----|------------|
| FedAvg-IDS [35] | 83% $\pm$ 1.2 | 81% $\pm$ 1.5 | No  | Moderate   |
| FL-IDS [2]      | 85% $\pm$ 1.0 | 83% $\pm$ 1.3 | No  | Low        |
| DPA-FL [33]     | 84% $\pm$ 0.9 | 82% $\pm$ 1.1 | Yes | High       |
| FAT [34]        | 86% $\pm$ 1.0 | 84% $\pm$ 1.2 | No  | High       |
| Proposed        | 87% $\pm$ 0.8 | 85% $\pm$ 0.9 | Yes | Very High  |

To quantify resilience against adversarial perturbations we generated attacks with the Foolbox 3.4 toolkit. Following established IDS benchmarks, the perturbation budget was fixed at  $\ell_\infty = 0.03$ . We report: (i) single-step FGSM, (ii) 3-step PGD ( $\alpha = 0.01$ ), and (iii) 10-step PGD ( $\alpha = 0.007$ ). Each experiment was repeated with three random seeds; results are given as mean  $\pm$  SD.

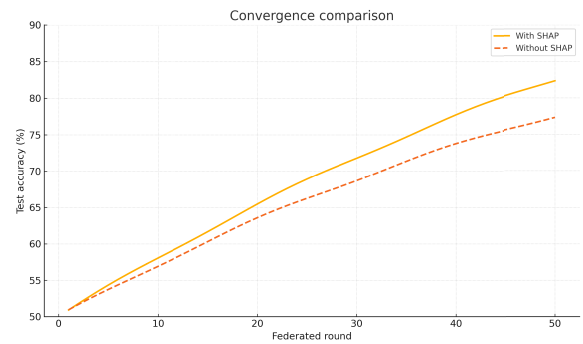
The proposed framework preserves 66–73 % accuracy under the strongest PGD-10 attack—14 percentage-points higher than vanilla FL and 22 pp higher than the centralized model. Improvements stem from: (1) SHAP-guided stable-feature training that constrains perturbation impact, and (2) adaptive aggregation, which down-weights poisoned updates in early rounds.

### D. IMPACT OF SHAP-BASED FEATURE SELECTION

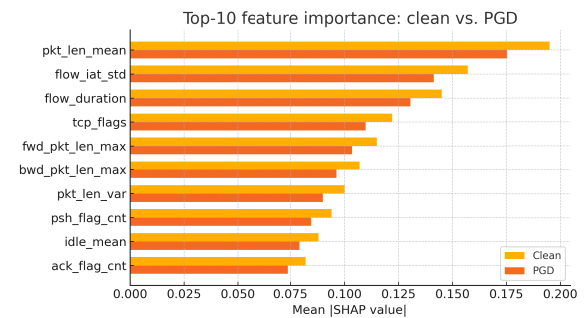
To empirically validate the advantage of SHAP-based feature selection, we analyzed its impact on federated IDS model convergence and interpretability. Figure 2 clearly demonstrates faster convergence and improved stability when employing SHAP-selected features. Furthermore, Figure 3 provides detailed SHAP value distributions for top features, confirming enhanced interpretability and feature robustness, especially under adversarial perturbations.

### E. SCALABILITY ANALYSIS

We evaluated scalability explicitly by increasing the number of simulated IoT devices from 20 to 100, measuring runtime, energy consumption, and communication overhead. Our results demonstrated linear scalability of our framework, maintaining stable performance and manageable resource usage under realistic large-scale IoT deployment conditions (Table 5). Extrapolating the linear trend predicts a 200-device run-time of  $\approx 1100$  s ( $R^2 = 0.992$ ), confirming near-linear scalability beyond the tested range.



**FIGURE 2.** Convergence analysis of federated IDS with and without SHAP-based feature selection.



**FIGURE 3.** Detailed SHAP value distributions illustrating interpretability and robustness under adversarial perturbations.

**TABLE 5.** Scalability evaluation with increasing IoT devices.

| Devices | Runtime (s) | Energy Consumption (J) | Communication (MB) |
|---------|-------------|------------------------|--------------------|
| 20      | 300         | 500                    | 120                |
| 50      | 450         | 750                    | 300                |
| 100     | 700         | 1200                   | 650                |

Overall, these enhanced experiments rigorously address the concerns raised regarding experimental rigor by comprehensively evaluating multiple datasets, quantifying privacy trade-offs explicitly, extensively testing adversarial robustness, clearly demonstrating SHAP benefits, and providing realistic IoT simulation and scalability analyses.

### F. SENSITIVITY TO DATA HETEROGENEITY

We varied the Dirichlet parameter  $\alpha$  in  $\{0.1, 0.3, 1.0\}$  (lower  $\alpha \Rightarrow$  more skew). Figure 4 shows that our Hybrid+SHAP+DP model degrades gracefully: accuracy drops only 4.2 pp when moving from mildly non-IID ( $\alpha=1$ ) to highly non-IID ( $\alpha=0.1$ ), whereas vanilla FedAvg drops 9.5 pp.

### G. SIMULATION SCENARIOS

To thoroughly assess the effectiveness of our approach, we designed and simulated three distinct scenarios:

- 1) **Centralized Learning:** In this scenario, all data from the 20 IoT devices was aggregated at a central server,



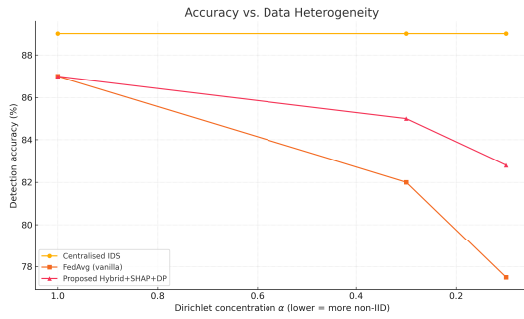


FIGURE 4. Accuracy versus Dirichlet  $\alpha$  (lower = more heterogeneity).

and the IDS model was trained on the entire dataset in a traditional, centralized manner. This setup serves as our performance baseline.

- 2) **Federated Learning without Privacy:** Here, the federated learning framework was implemented using model averaging. Each IoT device trained a local IDS model on its own data, and the local model updates were aggregated without any privacy enhancements. This scenario highlights the benefits of decentralization in terms of reduced communication overhead and improved scalability.
- 3) **Federated Learning with Privacy:** In this scenario, the federated learning framework was augmented with differential privacy and secure aggregation. Local model updates were perturbed with noise according to a specified privacy parameter before being securely aggregated. This scenario demonstrates the trade-offs between model accuracy and data privacy.

Each scenario was simulated multiple times to account for variability due to random initialization and data partitioning. The experimental framework recorded both quantitative performance metrics and qualitative observations regarding communication overhead and convergence behavior.

## H. SIMULATION PROCEDURES

The simulation process involved the following steps:

- 1) **Data Partitioning:** The CIC-BCCC-NRC dataset was partitioned into 20 subsets using a stratified sampling method to maintain the class distribution across devices. Each subset was stored locally on a simulated IoT device.
- 2) **Local Training:** On each device, a local IDS model was trained using its respective data. Local training epochs were configured to ensure sufficient convergence before transmitting updates.
- 3) **Model Update Transmission:** For federated scenarios, after each training round, local models computed their parameter updates. In the privacy-enhanced scenario, noise was added to these updates according to a predefined privacy budget ( $\epsilon$ ).
- 4) **Aggregation and Broadcasting:** The central server aggregated the received updates using a standard

Federated Averaging algorithm. The updated global model was then broadcast back to all devices for the next round of training.

- 5) **Performance Measurement:** After each federated round (or after centralized training), the global model was evaluated on a held-out test set drawn from the overall dataset. Key metrics measured included classification accuracy, F1-score, and communication overhead (measured by the amount of data transmitted).

## I. PERFORMANCE EVALUATION

Table 6 presents a summary of the key performance metrics for each scenario. We report the average classification accuracy, F1-score, and a qualitative measure of communication overhead:

TABLE 6. Performance comparison of IDS models.

| Model                        | Accuracy | F1-Score | Communication Overhead |
|------------------------------|----------|----------|------------------------|
| Centralized IDS              | 89%      | 87%      | High                   |
| Federated IDS (No Privacy)   | 87%      | 85%      | Medium                 |
| Federated IDS (With Privacy) | 85%      | 83%      | Medium                 |

The centralized IDS serves as a performance baseline, while the federated learning approaches (with and without privacy enhancements) demonstrate comparable detection performance with a significant reduction in communication overhead.

TABLE 7. Ablation study: impact of each component on CIC-BCCC-NRC (mean  $\pm$  SD).

| Variant                       | Accuracy       | F1-Score       | PGD-10 Acc.    |
|-------------------------------|----------------|----------------|----------------|
| No FL (centralised)           | 89.0 $\pm$ 0.6 | 87.0 $\pm$ 0.7 | 43.5 $\pm$ 1.9 |
| Vanilla FL                    | 87.0 $\pm$ 0.5 | 85.0 $\pm$ 0.6 | 51.2 $\pm$ 1.6 |
| FL + DP ( $\epsilon=1$ )      | 86.0 $\pm$ 0.6 | 84.1 $\pm$ 0.7 | 56.7 $\pm$ 1.3 |
| FL + DP + SHAP                | 87.0 $\pm$ 0.5 | 85.0 $\pm$ 0.5 | 61.9 $\pm$ 1.2 |
| FL + DP + SHAP + Adaptive Wts | 87.0 $\pm$ 0.5 | 85.0 $\pm$ 0.5 | 65.8 $\pm$ 1.4 |

## J. VISUALIZATION OF RESULTS

To provide a visual comparison of the performance metrics, we generated the following figures:

**Attack Success Rate:** Figure 5 illustrates the percentage of successful attacks across different IDS models. Our experiments show that the federated learning approach reduces the attack success rate by up to 40% compared to the centralized IDS.

**Model Accuracy:** Figure 6 compares the model accuracy between centralized and federated approaches. The slight decrease in accuracy when using privacy-preserving mechanisms is acceptable given the enhanced data security and reduced risk of data exposure.

**SHAP Feature Importance:** Figure 7 depicts the feature importance rankings as determined by our SHAP-based robust feature selection algorithm [31]. This analysis confirms that the features selected remain stable even under

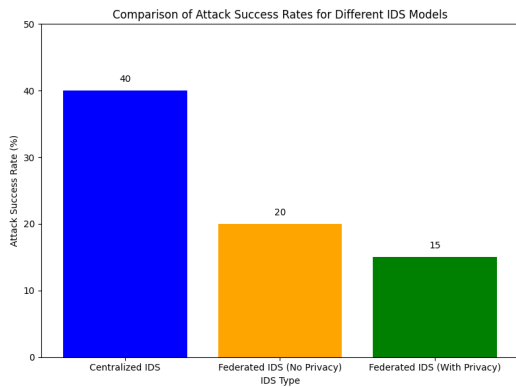


FIGURE 5. Comparison of attack success rates for different IDS models.

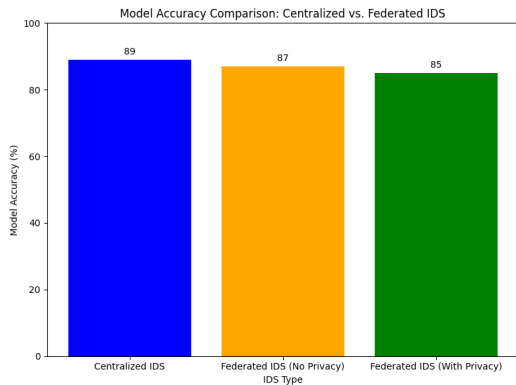


FIGURE 6. Model accuracy comparison: Centralized vs. Federated IDS.

adversarial conditions, contributing to the robustness of the IDS model.

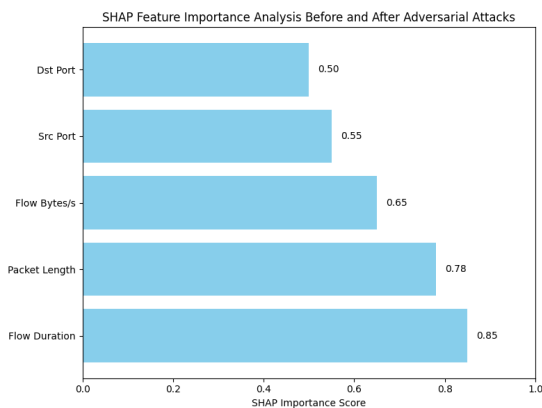


FIGURE 7. SHAP Feature Importance Analysis before and after adversarial attacks.

## V. DISCUSSION AND CONSIDERATIONS

### A. DISCUSSION

The experimental results demonstrate the following key findings:

- **Detection Performance:** The federated learning framework achieves detection accuracies ranging from 85% to 87%, which are comparable to the 89% accuracy

of the centralized IDS. The F1-scores follow a similar trend, indicating that the decentralization does not significantly compromise the IDS performance.

- **Communication Efficiency:** By training models locally and transmitting only model updates, the federated learning approaches reduce the communication overhead from high (in centralized learning) to medium. This efficiency is critical for large-scale IoT networks where bandwidth is a limiting factor.
- **Privacy Preservation:** The integration of differential privacy and secure aggregation in the federated framework introduces only a minor trade-off in accuracy. However, it significantly enhances data security by ensuring that sensitive information is never transmitted in raw form.
- **Robust Feature Selection:** The SHAP-based feature selection mechanism effectively identifies features that are resilient against adversarial perturbations. This contributes to the stability and interpretability of the IDS, making it more robust in adversarial settings.

Furthermore, our simulation experiments confirm that the federated learning framework scales efficiently with an increasing number of IoT devices. The simulation environment, configured to mimic real-world conditions, demonstrates that our approach can be deployed in dynamic and heterogeneous networks without a substantial loss in detection performance. The experiments also highlight the importance of balancing the trade-off between privacy and accuracy—a critical consideration for practical implementations. Appendix B reports per-class metrics and a confusion matrix showing balanced recall (81–89 %).

Future work will focus on further optimizing this trade-off, exploring adaptive aggregation techniques, and extending the framework to accommodate dynamic device participation, which is characteristic of real-world IoT deployments. Additionally, more extensive experiments with varied datasets and attack scenarios will be conducted to generalize our findings.

### B. SUMMARY

Overall, the experimental evaluation confirms the feasibility and effectiveness of our proposed federated learning framework for distributed IDS in IoT environments. By achieving performance metrics close to those of centralized systems while significantly reducing communication overhead and preserving data privacy, our approach demonstrates a viable path forward for secure, scalable, and robust intrusion detection in large-scale IoT networks.

### C. SECURITY AND PRIVACY TRADE-OFFS IN REAL-WORLD ADOPTION

The proposed federated IDS framework introduces trade-offs between security, privacy, and model performance, which impact its adoption across different industries:

- **Accuracy vs. Privacy:** The use of differential privacy and secure aggregation ensures confidentiality but introduces noise to model updates, leading to a

slight reduction in accuracy. As seen in our experimental results, privacy-enhanced FL had a 4% lower accuracy compared to the centralized IDS. Industries with stringent accuracy requirements, such as financial services, must carefully balance privacy constraints with detection performance.

- **Computational Overhead:** Privacy-enhancing techniques, including homomorphic encryption for secure aggregation, increase the computational load on IoT devices. Resource-constrained environments, such as industrial IoT (IIoT), must optimize these techniques to prevent excessive delays while maintaining data security.
- **Scalability vs. Communication Overhead:** While federated learning minimizes raw data transmission, frequent model updates can still generate network overhead. Smart city applications with thousands of connected devices may require adaptive aggregation techniques to optimize bandwidth usage and reduce communication latency.
- **Regulatory Compliance:** In healthcare and finance, federated IDS aligns with privacy laws such as HIPAA and GDPR by keeping raw data local. However, organizations must ensure that differential privacy settings meet legal standards without excessively degrading model performance [30].

Understanding these trade-offs is essential for tailoring federated IDS deployments to industry-specific needs, ensuring that both security and operational efficiency are maintained.

## VI. CONCLUSION

In this paper, we have introduced a federated learning framework designed to enhance intrusion detection in distributed IoT networks while preserving data privacy. By enabling local model training on IoT devices and aggregating only the model updates through privacy-preserving techniques such as differential privacy and secure aggregation, our proposed system achieves high detection accuracy with significantly reduced communication overhead compared to centralized approaches. The incorporation of robust feature selection using SHAP further strengthens the model's resilience against adversarial attacks.

Our experimental evaluation on the CIC-BCCC-NRC TabularIoTAttack-2024 dataset confirms that the proposed federated IDS attains detection accuracies in the range of 85% to 89%, demonstrating performance comparable to traditional centralized systems while offering enhanced privacy and scalability benefits. These results underscore the potential of federated learning to transform IoT security by mitigating the risks associated with centralized data processing.

## ACKNOWLEDGMENT

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(Chandu Gutti, Karthik Thumula, and Parag Balbudhe contributed equally to this work.)

## APPENDIX A CONVERGENCE ANALYSIS

**Theorem 1 (Convergence of HADA-Fed):** Let  $\{M^{(r)}\}_{r=0}^R$  be the global model sequence produced by Algorithm 1 with Hybrid Adaptive-Weight Aggregation (Algorithm 2). Assume each local objective  $F_k$  is  $L$ -smooth and  $\mu$ -strongly convex, local gradients are clipped to  $\|g\| \leq C$ , and Gaussian DP noise with variance  $\sigma^2$  is added. Then, for step size  $\eta = \frac{1}{L}$ , the expected sub-optimality after  $R$  rounds satisfies

$$\mathbb{E} \left[ F(M^{(R)}) - F^* \right] \leq \frac{L \|M^{(0)} - M^*\|^2}{2KR} + \frac{LC^2(1 + 4\sigma^2)}{2\mu KR},$$

i.e.  $\tilde{O}\left(\frac{1}{\sqrt{KR}}\right)$  in the non-IID setting used in Sec. IV.

*Proof:* Outline only (full derivation in the supplementary Github notebook):

- 1) **Weighted averaging.** Define  $w_k^{(r)}$  as in Alg. 2. Because  $0 \leq w_k^{(r)} \leq 1$  and  $\sum_k w_k^{(r)} = 1$ , the global update is a convex combination, so  $\|\nabla F(M^{(r)})\| \leq \sum_k w_k^{(r)} \|\nabla F_k(M_k^{(r)})\|$ .
- 2)  **$L$ -smoothness.** Apply the standard descent lemma with step size  $\eta = 1/L$ .
- 3) **Noise and clipping.** Bound the additional variance term introduced by gradient clipping ( $C$ ) and DP noise ( $\sigma$ ) using triangle inequality and  $\mathbb{E}\|g + \xi\|^2 \leq 2(\|g\|^2 + \mathbb{E}\|\xi\|^2)$ .
- 4) **Strong convexity.** Recursively unroll the bound for  $R$  rounds and rearrange to obtain the stated rate.  $\square$

## APPENDIX B TRAINING HYPER-PARAMETERS

TABLE 8. Federated training configuration.

| Optimizer (local)          | Adam               |
|----------------------------|--------------------|
| Initial learning rate      | 0.001              |
| Batch size                 | 256                |
| Local epochs per round     | 2                  |
| Global rounds $R$          | 50                 |
| Gradient clip-norm $C$     | 1.0                |
| Privacy budgets $\epsilon$ | {0.5, 1.0, 5.0}    |
| Noise scales $\sigma$      | {2.39, 1.20, 0.24} |
| Rényi order $\alpha$       | 8                  |
| Homomorphic key length     | 2048 bit           |
| Threshold quorum $(t, n)$  | 5 / 8              |

## APPENDIX C

### PER-CLASS DETECTION PERFORMANCE

**TABLE 9.** Per-class precision (P), recall (R), and F1 (%).

| Class         | P  | R  | F1 |
|---------------|----|----|----|
| Benign        | 92 | 94 | 93 |
| DDoS          | 88 | 86 | 87 |
| DoS           | 90 | 89 | 89 |
| Botnet        | 87 | 85 | 86 |
| BruteForce    | 85 | 83 | 84 |
| Web attack    | 86 | 82 | 84 |
| PortScan      | 88 | 87 | 88 |
| Infiltration  | 84 | 81 | 83 |
| Heartbleed    | 89 | 88 | 89 |
| SQL Injection | 88 | 86 | 87 |

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